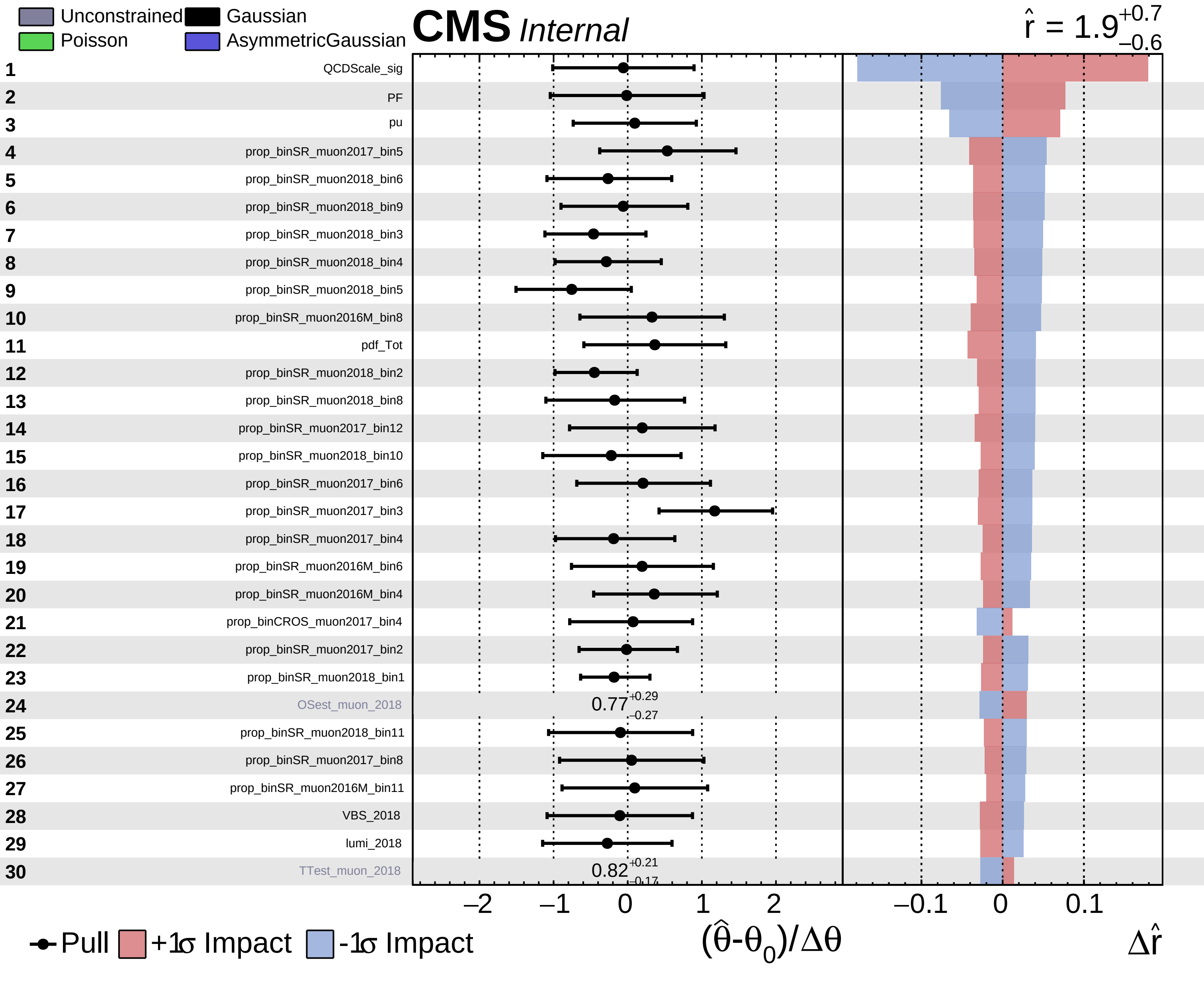
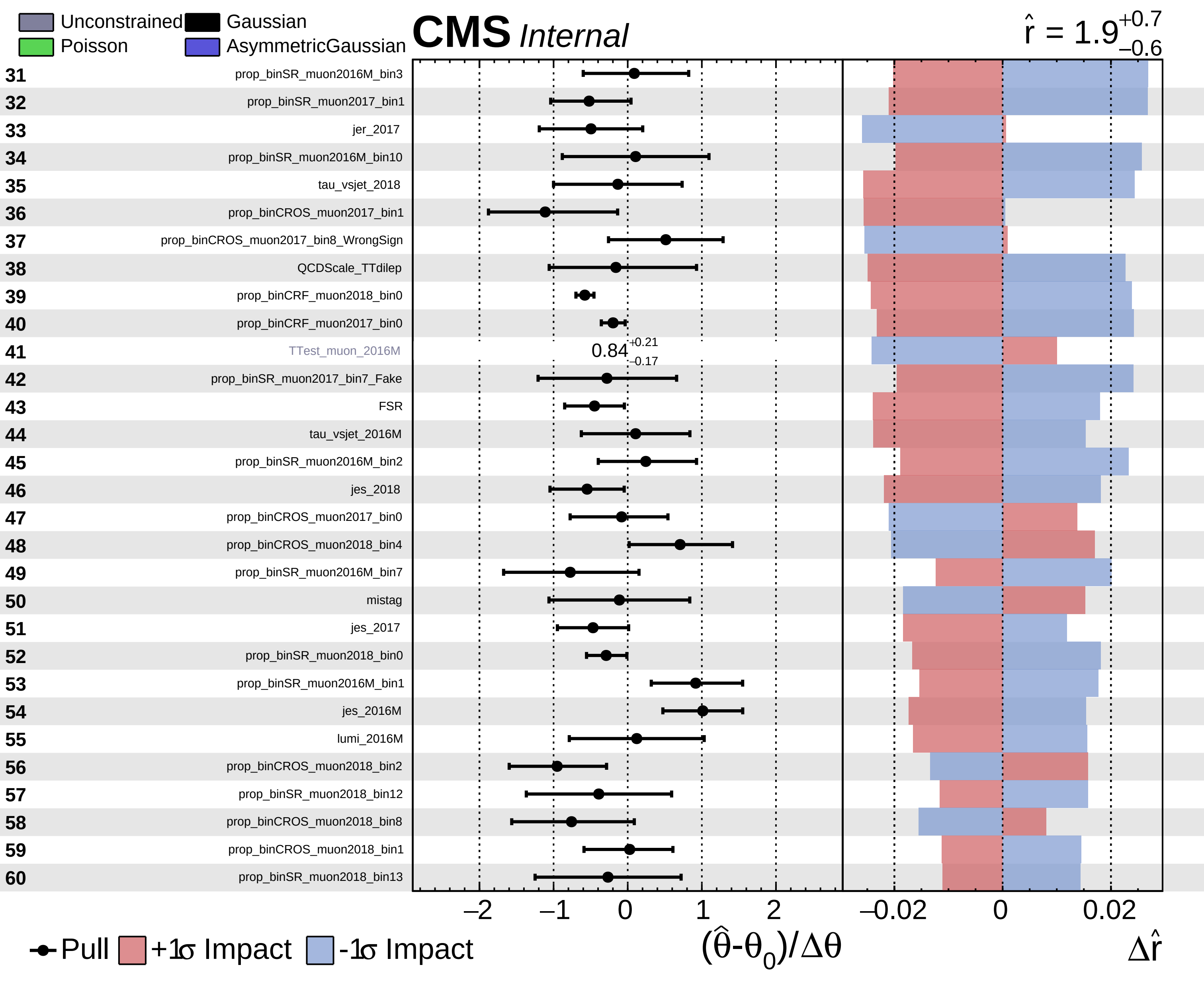
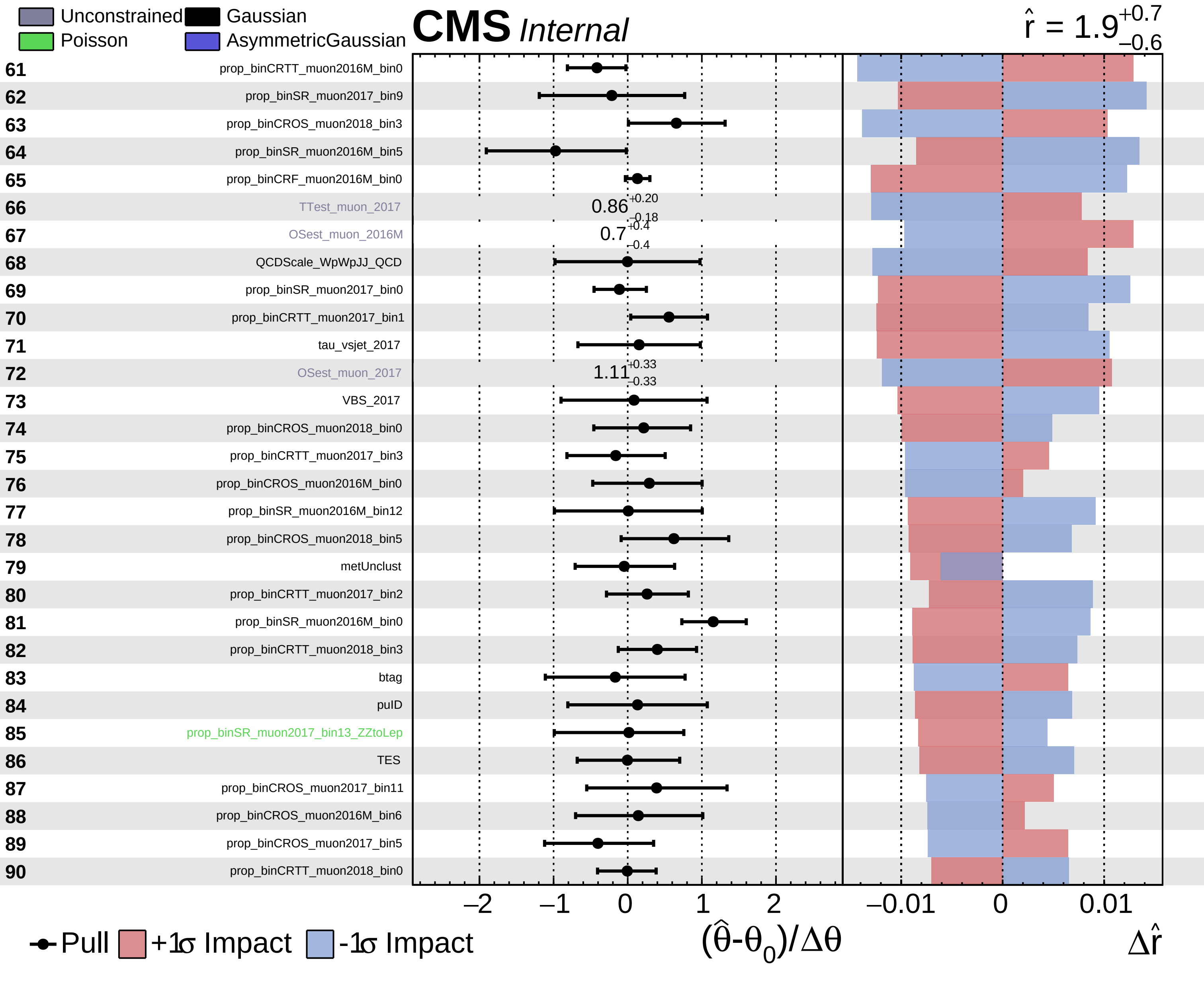
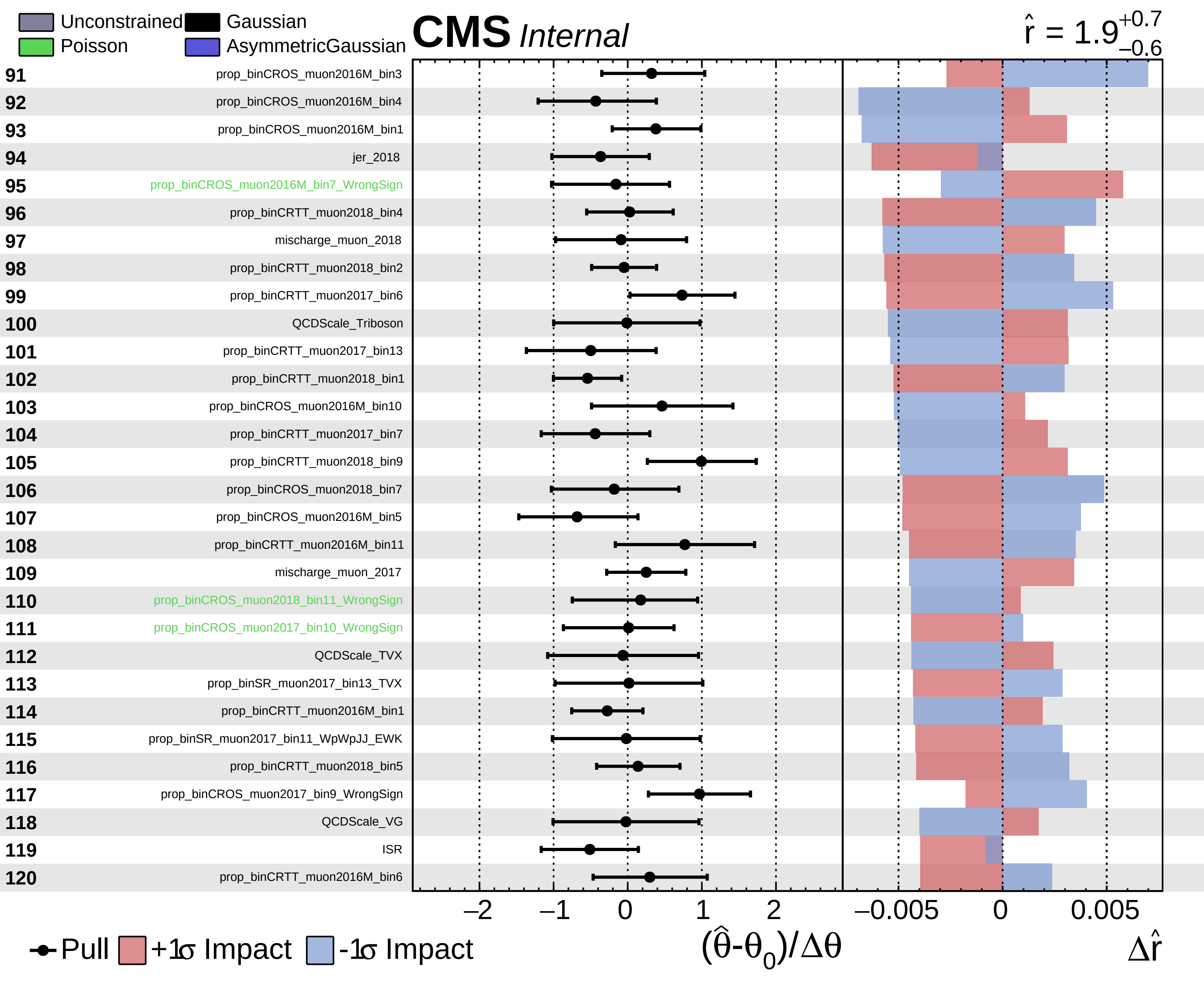


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.9^{+0.7}_{-0.6}$
1 QCDScale_sig
2 PF
3 pu
4 prop_binSR_muon2017_bin5
5 prop_binSR_muon2018_bin6
6 prop_binSR_muon2018_bin9
7 prop_binSR_muon2018_bin3
8 prop_binSR_muon2018_bin4
9 prop_binSR_muon2018_bin5
10 prop_binSR_muon2016M_bin8
11 pdf_Tot
12 prop_binSR_muon2018_bin2
13 prop_binSR_muon2018_bin8
14 prop_binSR_muon2017_bin12
15 prop_binSR_muon2018_bin10
16 prop_binSR_muon2017_bin6
17 prop_binSR_muon2017_bin3
18 prop_binSR_muon2017_bin4
19 prop_binSR_muon2016M_bin6
20 prop_binSR_muon2016M_bin4
21 prop_binCROS_muon2017_bin4
22 prop_binSR_muon2017_bin2
23 prop_binSR_muon2018_bin1
24 OSest_muon_2018
$0.77^{+0.29}_{-0.27}$
25 prop_binSR_muon2018_bin11
26 prop_binSR_muon2017_bin8
27 prop_binSR_muon2016M_bin11
28 VBS_2018
29 lumi_2018
30 TTest_muon_2018
$0.82^{+0.21}_{-0.17}$
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

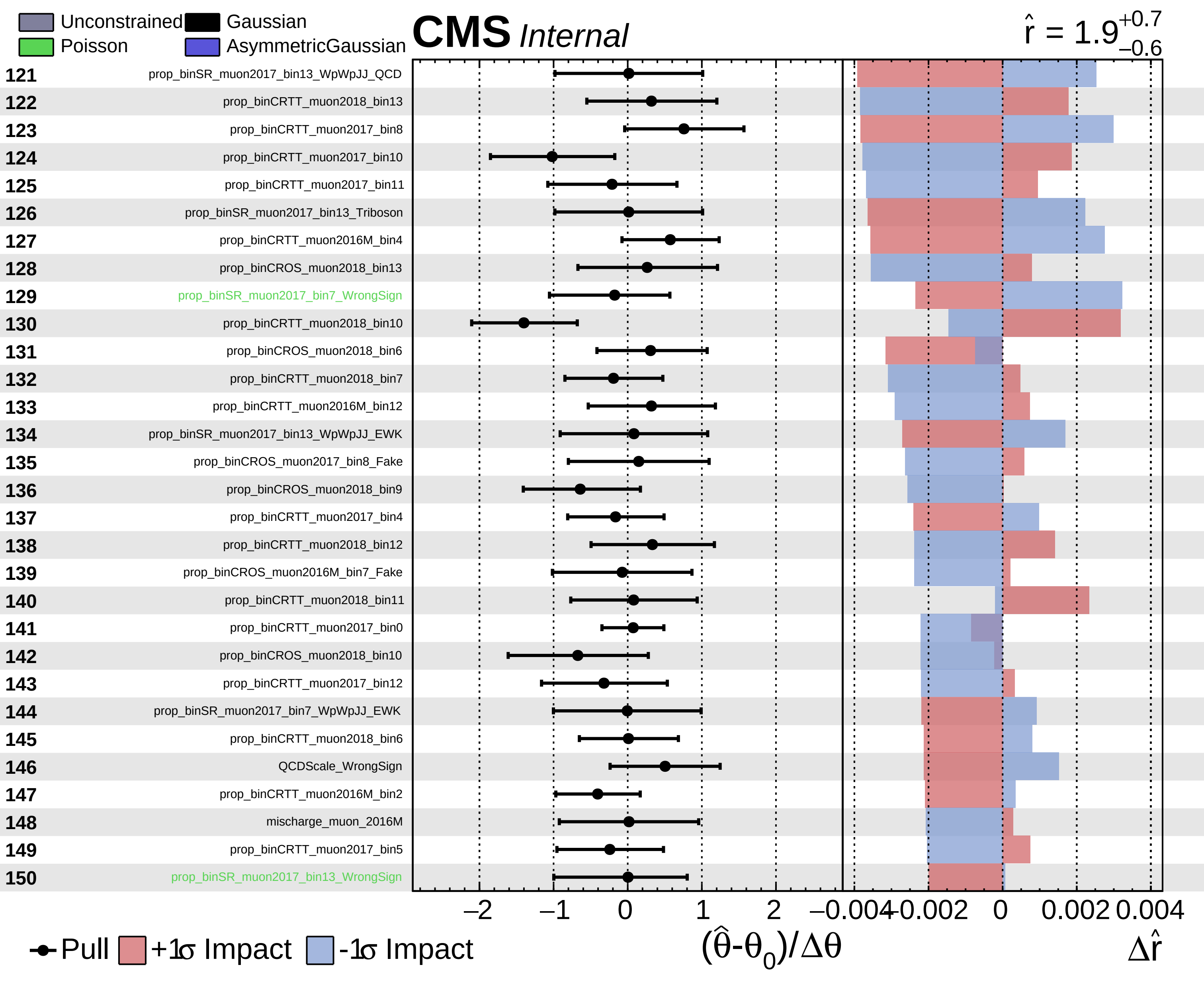


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.9 +0.7 −0.6
31 prop_binSR_muon2016M_bin3
32 prop_binSR_muon2017_bin1
33 jer_2017
34 prop_binSR_muon2016M_bin10
35 tau_vsjet_2018
36 prop_binCROS_muon2017_bin1
37 prop_binCROS_muon2017_bin8_WrongSign
38 QCDScale_TTdilep
39 prop_binCRF_muon2018_bin0
40 prop_binCRF_muon2017_bin0
41 TTest_muon_2016M
0.84 +0.21 −0.17
42 prop_binSR_muon2017_bin7_Fake
43 FSR
44 tau_vsjet_2016M
45 prop_binSR_muon2016M_bin2
46 jes_2018
47 prop_binCROS_muon2017_bin0
48 prop_binCROS_muon2018_bin4
49 prop_binSR_muon2016M_bin7
50 mistag
51 jes_2017
52 prop_binSR_muon2018_bin0
53 prop_binSR_muon2016M_bin1
54 jes_2016M
55 lumi_2016M
56 prop_binCROS_muon2018_bin2
57 prop_binSR_muon2018_bin12
58 prop_binCROS_muon2018_bin8
59 prop_binCROS_muon2018_bin1
60 prop_binSR_muon2018_bin13
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

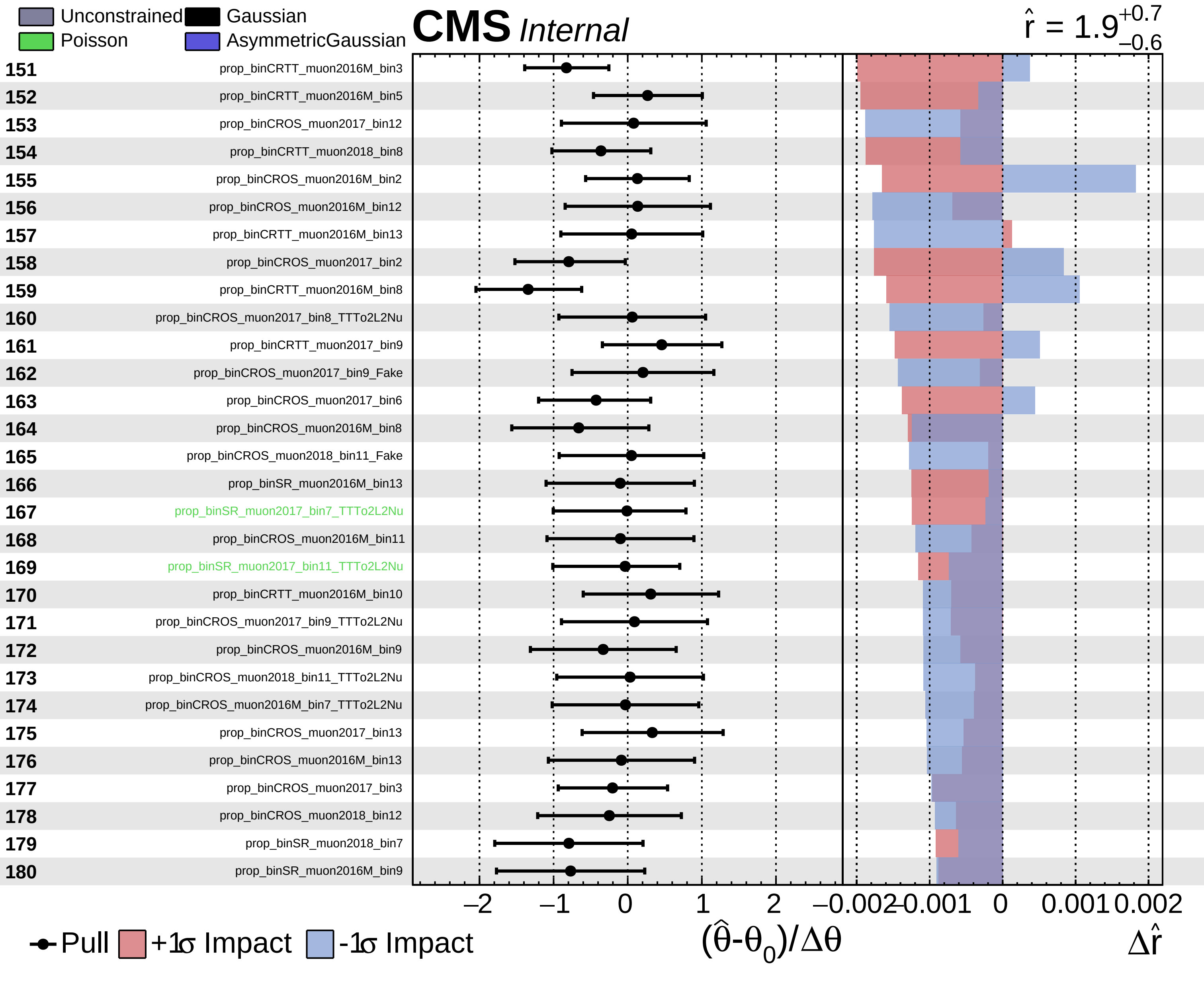

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.9^{+0.7}_{-0.6}$
61 prop_binCRTT_muon2016M_bin0
62 prop_binSR_muon2017_bin9
63 prop_binCROS_muon2018_bin3
64 prop_binSR_muon2016M_bin5
65 prop_binCRF_muon2016M_bin0
66 TTest_muon_2017 $0.86^{+0.20}_{-0.18}$
67 OSest_muon_2016M $0.7^{+0.4}_{-0.4}$
68 QCDScale_WpWpJJ_QCD
69 prop_binSR_muon2017_bin0
70 prop_binCRTT_muon2017_bin1
71 tau_vsjet_2017
72 OSest_muon_2017 $1.11^{+0.33}_{-0.33}$
73 VBS_2017
74 prop_binCROS_muon2018_bin0
75 prop_binCRTT_muon2017_bin3
76 prop_binCROS_muon2016M_bin0
77 prop_binSR_muon2016M_bin12
78 prop_binCROS_muon2018_bin5
79 metUnclust
80 prop_binCRTT_muon2017_bin2
81 prop_binSR_muon2016M_bin0
82 prop_binCRTT_muon2018_bin3
83 btag
84 puID
85 prop_binSR_muon2017_bin13_ZZtoLep
86 TES
87 prop_binCROS_muon2017_bin11
88 prop_binCROS_muon2016M_bin6
89 prop_binCROS_muon2017_bin5
90 prop_binCRTT_muon2018_bin0
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



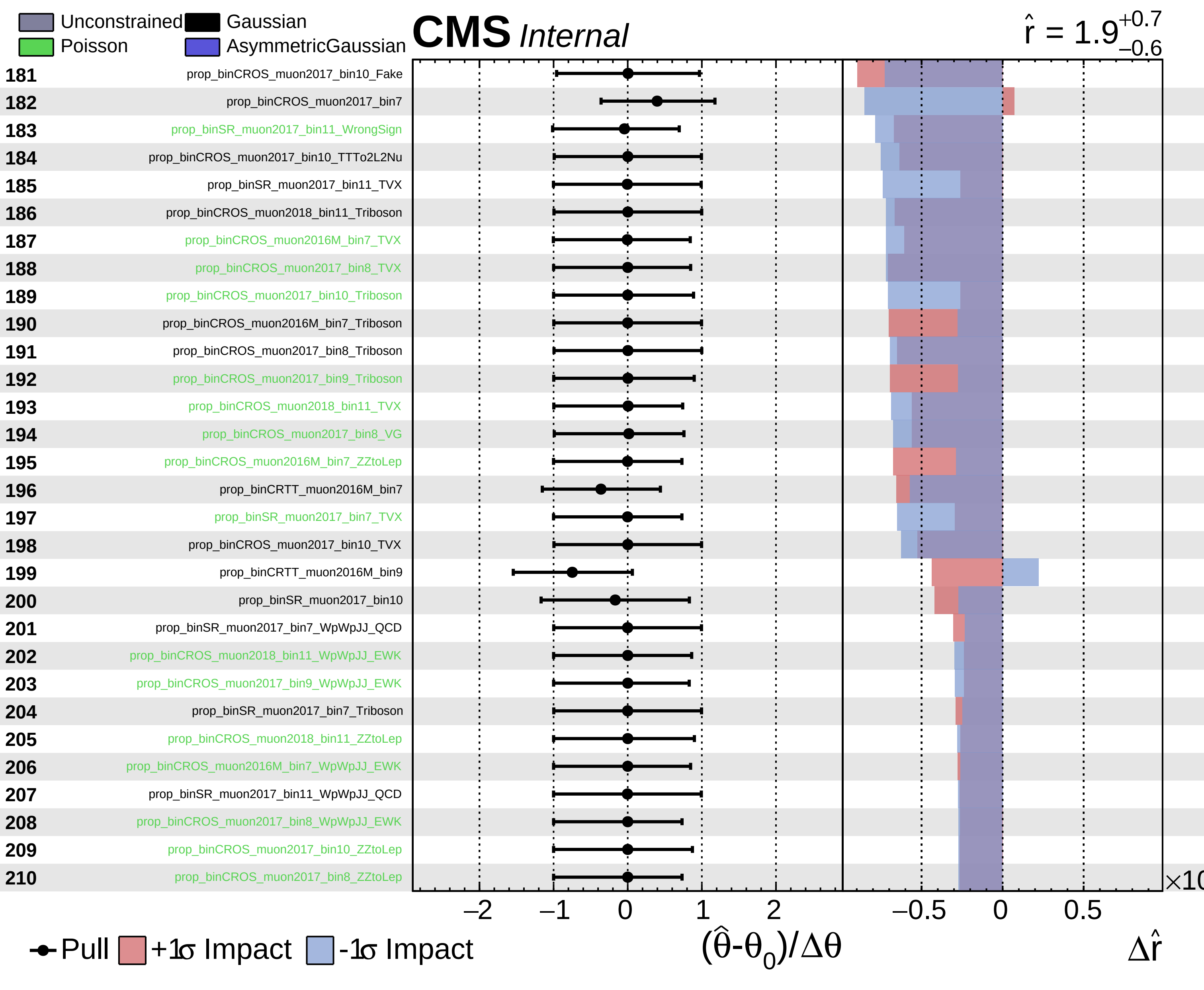
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.9^{+0.7}_{-0.6}$
91 prop_binCROS_muon2016M_bin3
92 prop_binCROS_muon2016M_bin4
93 prop_binCROS_muon2016M_bin1
94 jer_2018
95 prop_binCROS_muon2016M_bin7_WrongSign
96 prop_binCRTT_muon2018_bin4
97 mischarge_muon_2018
98 prop_binCRTT_muon2018_bin2
99 prop_binCRTT_muon2017_bin6
100 QCDScale_Triboson
101 prop_binCRTT_muon2017_bin13
102 prop_binCRTT_muon2018_bin1
103 prop_binCROS_muon2016M_bin10
104 prop_binCRTT_muon2017_bin7
105 prop_binCRTT_muon2018_bin9
106 prop_binCROS_muon2018_bin7
107 prop_binCROS_muon2016M_bin5
108 prop_binCRTT_muon2016M_bin11
109 mischarge_muon_2017
110 prop_binCROS_muon2018_bin11_WrongSign
111 prop_binCROS_muon2017_bin10_WrongSign
112 QCDScale_TVX
113 prop_binSR_muon2017_bin13_TVX
114 prop_binCRTT_muon2016M_bin1
115 prop_binSR_muon2017_bin11_WpWpJJ_EWK
116 prop_binCRTT_muon2018_bin5
117 prop_binCROS_muon2017_bin9_WrongSign
118 QCDScale_VG
119 ISR
120 prop_binCRTT_muon2016M_bin6
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



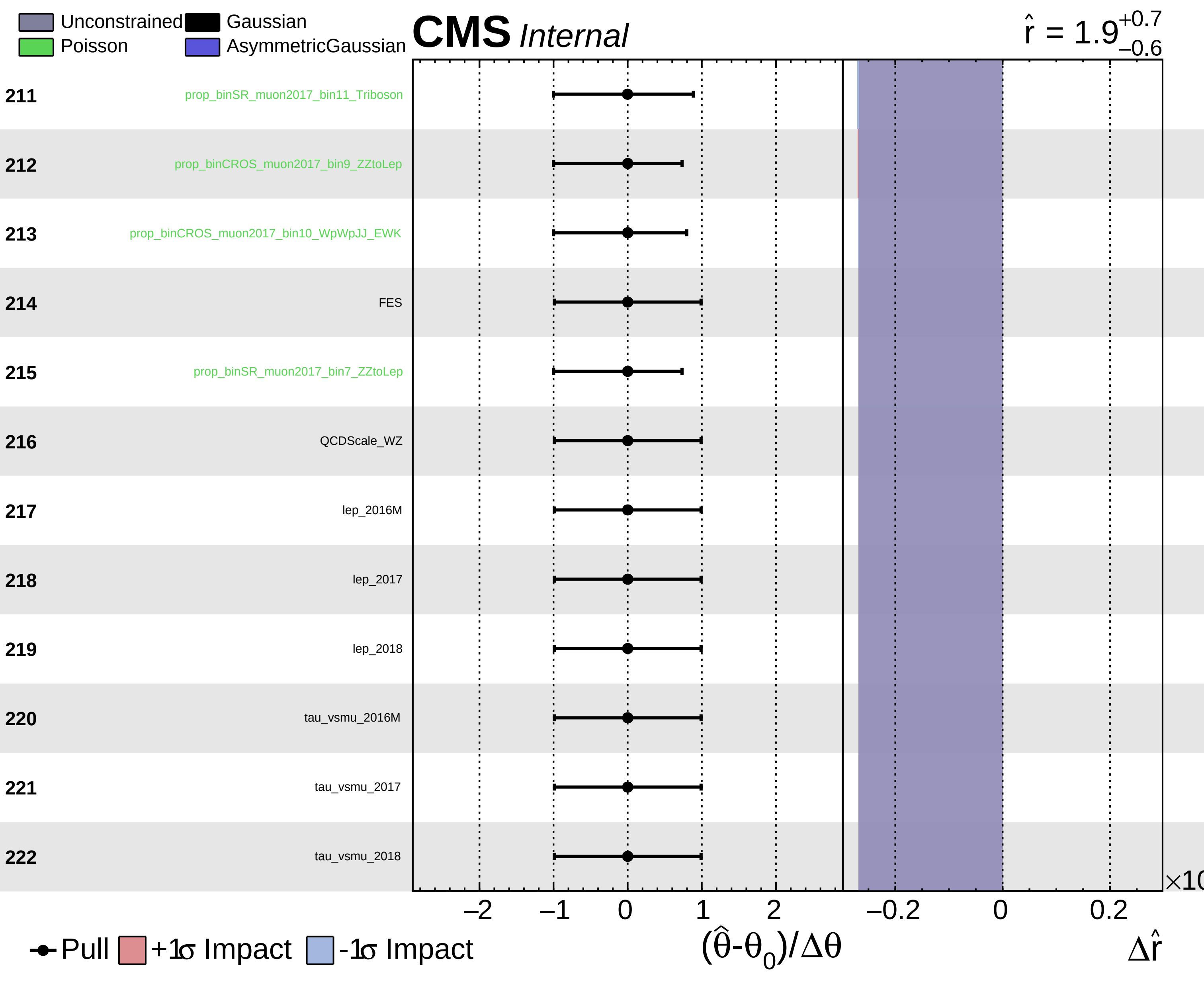
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.9^{+0.7}_{-0.6}$
121 prop_binSR_muon2017_bin13_WpWpJJ_QCD
122 prop_binCRTT_muon2018_bin13
123 prop_binCRTT_muon2017_bin8
124 prop_binCRTT_muon2017_bin10
125 prop_binCRTT_muon2017_bin11
126 prop_binSR_muon2017_bin13_Triboson
127 prop_binCRTT_muon2016M_bin4
128 prop_binCROS_muon2018_bin13
129 prop_binSR_muon2017_bin7_WrongSign
130 prop_binCRTT_muon2018_bin10
131 prop_binCROS_muon2018_bin6
132 prop_binCRTT_muon2018_bin7
133 prop_binCRTT_muon2016M_bin12
134 prop_binSR_muon2017_bin13_WpWpJJ_EWK
135 prop_binCROS_muon2017_bin8_Fake
136 prop_binCROS_muon2018_bin9
137 prop_binCRTT_muon2017_bin4
138 prop_binCRTT_muon2018_bin12
139 prop_binCROS_muon2016M_bin7_Fake
140 prop_binCRTT_muon2018_bin11
141 prop_binCRTT_muon2017_bin0
142 prop_binCROS_muon2018_bin10
143 prop_binCRTT_muon2017_bin12
144 prop_binSR_muon2017_bin7_WpWpJJ_EWK
145 prop_binCRTT_muon2018_bin6
146 QCDScale_WrongSign
147 prop_binCRTT_muon2016M_bin2
148 mischarge_muon_2016M
149 prop_binCRTT_muon2017_bin5
150 prop_binSR_muon2017_bin13_WrongSign
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.9^{+0.7}_{-0.6}$
151 prop_binCRTT_muon2016M_bin3
152 prop_binCRTT_muon2016M_bin5
153 prop_binCROS_muon2017_bin12
154 prop_binCRTT_muon2018_bin8
155 prop_binCROS_muon2016M_bin2
156 prop_binCROS_muon2016M_bin12
157 prop_binCRTT_muon2016M_bin13
158 prop_binCROS_muon2017_bin2
159 prop_binCRTT_muon2016M_bin8
160 prop_binCROS_muon2017_bin8_TTTo2L2Nu
161 prop_binCRTT_muon2017_bin9
162 prop_binCROS_muon2017_bin9_Fake
163 prop_binCROS_muon2017_bin6
164 prop_binCROS_muon2016M_bin8
165 prop_binCROS_muon2018_bin11_Fake
166 prop_binSR_muon2016M_bin13
167 prop_binSR_muon2017_bin7_TTTo2L2Nu
168 prop_binCROS_muon2016M_bin11
169 prop_binSR_muon2017_bin11_TTTo2L2Nu
170 prop_binCRTT_muon2016M_bin10
171 prop_binCROS_muon2017_bin9_TTTo2L2Nu
172 prop_binCROS_muon2016M_bin9
173 prop_binCROS_muon2018_bin11_TTTo2L2Nu
174 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
175 prop_binCROS_muon2017_bin13
176 prop_binCROS_muon2016M_bin13
177 prop_binCROS_muon2017_bin3
178 prop_binCROS_muon2018_bin12
179 prop_binSR_muon2018_bin7
180 prop_binSR_muon2016M_bin9
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.9^{+0.7}_{-0.6}$
181 prop_binCROS_muon2017_bin10_Fake
182 prop_binCROS_muon2017_bin7
183 prop_binSR_muon2017_bin11_WrongSign
184 prop_binCROS_muon2017_bin10_TTTo2L2Nu
185 prop_binSR_muon2017_bin11_TVX
186 prop_binCROS_muon2018_bin11_Triboson
187 prop_binCROS_muon2016M_bin7_TVX
188 prop_binCROS_muon2017_bin8_TVX
189 prop_binCROS_muon2017_bin10_Triboson
190 prop_binCROS_muon2016M_bin7_Triboson
191 prop_binCROS_muon2017_bin8_Triboson
192 prop_binCROS_muon2017_bin9_Triboson
193 prop_binCROS_muon2018_bin11_TVX
194 prop_binCROS_muon2017_bin8_VG
195 prop_binCROS_muon2016M_bin7_ZZtoLep
196 prop_binCRTT_muon2016M_bin7
197 prop_binSR_muon2017_bin7_TVX
198 prop_binCROS_muon2017_bin10_TVX
199 prop_binCRTT_muon2016M_bin9
200 prop_binSR_muon2017_bin10
201 prop_binSR_muon2017_bin7_WpWpJJ_QCD
202 prop_binCROS_muon2018_bin11_WpWpJJ_EWK
203 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
204 prop_binSR_muon2017_bin7_Triboson
205 prop_binCROS_muon2018_bin11_ZZtoLep
206 prop_binCROS_muon2016M_bin7_WpWpJJ_EWK
207 prop_binSR_muon2017_bin11_WpWpJJ_QCD
208 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
209 prop_binCROS_muon2017_bin10_ZZtoLep
210 prop_binCROS_muon2017_bin8_ZZtoLep
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.9^{+0.7}_{-0.6}$
211 prop_binSR_muon2017_bin11_Triboson
212 prop_binCROS_muon2017_bin9_ZZtoLep
213 prop_binCROS_muon2017_bin10_WpWpJJ_EWK
214 FES
215 prop_binSR_muon2017_bin7_ZZtoLep
216 QCDScale_WZ
217 lep_2016M
218 lep_2017
219 lep_2018
220 tau_vsmu_2016M
221 tau_vsmu_2017
222 tau_vsmu_2018
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$